Modeling and Visualizing Performance of Professional Soccer Goalkeepers

By

Noah Kan

CAPSTONE PROJECT

Submitted in partial satisfaction of the requirements for the degree of

BACHELOR OF SCIENCE

in

STATISTICS AND DATA SCIENCE

in the

COLLEGE OF SCIENCE

at

CALIFORNIA STATE UNIVERSITY,
MONTEREY BAY

Approved:

(Advisor)

Dr. Steven Kim

Department of Mathematics and Statistics

Spring 2026

Acknowledgements

I would like to express my sincere gratitude to Dr. Kim for his guidance throughout my research projects and coursework. His patience and support allowed me to grow not only as a student, but also as a person. I am also deeply thankful to Dr. Canner for providing me with numerous opportunities and for creating such a welcoming and supportive environment at CSUMB, where I was able to thrive in ways I could not have elsewhere. I would like to thank Dr. Unfried as well for her rigorous expectations and critical feedback on my writing, which challenged me to become a more disciplined and thoughtful student. Finally, I am incredibly grateful to my parents and friends for their constant encouragement and for standing by my side throughout this journey.

Abstract

The goalkeeper on the soccer pitch has been historically one of the most important players with a direct impact on their teams success. There is previous research on goalkeepers' performance but very little research on their evaluation based on shots coming from different parts of the field. In this project, we addressed four objectives: (1) testing and estimating the probability of a goal by distance and angle of a shot among K League players, (2) assessing the reliability of goalkeepers' performance measure by comparing the save percentages between two seasons, (3) comparing the proportion of shots attempted in the box by the proportion of time when a team is leading or trailing and (4) evaluating whether goalkeepers' ages affects their performance. We used generalized linear models and generalized linear mixed-effects models to answer each research question. For each objective, we found that (1) the the distance and angle of a shot are statistically significant factors fir the probability of a goal, and heatmaps are useful to graphically evaluate the goalkeepers; (2) the correlation of save percentages between two seasons is 0.5788 ($p = 0.044$) ; (3) field players attempt more shots inside the box when the team is leading, but the relationship was not statistically significant ($p = 0.15874$); and (4) the goalkeepers' performance is not explained by their ages ($p = 0.78948$).

Contents

Acknowledgements ii

Abstract iii

1 Introduction 1

2 Methods 3

2.1 Data Collection . . . 3

2.2 Statistical Methods . . . 4

2.2.1 First Objective . . . 4

2.2.2 Second Objective . . . 5

2.2.3 Third Objective . . . 5

2.2.4 Fourth Objective . . . 6

3 Results 7

3.1 First Objective . . . 7

3.2 Second Objective . . . 9

3.3 Third Objective . . . 10

3.4 Fourth Objective . . . 11

4 Discussion 12

4.1 Interpreting Results . . . 12

4.2 Limitations . . . 12

4.3 Challenges and Future Work . . . 13

5 Conclusion 15

1 Introduction

Goalkeeper analysis is very important in soccer, as goalkeepers are directly related to winning. It was discovered that the save percentage is directly associated with the team's probability of winning (Christiansen & Houser, 2024). This study highlights the importance of a goalkeeper reaching their maximum potential to impact team success, and also provides information about the likelihood of a goal based on shot location from the attacker's perspective.

Previous research examined how goalkeepers defend various areas of the goal across a range of in-game situations. Training programs can create more effective recommendations and strategies for goalkeepers to react where the balls are targeted on goal (Park, Choi, Bang, & Park, 2016). However, goalkeepers have not been graphically evaluated by distance and angle from the goal which is a focus of this project. Studying the high volume areas that players attempt shots was useful in goalkeeper training. Creating situational training scenarios in which goalkeepers attempt to predict shot direction before the ball is struck can lead to more successful saves. In addition to identifying the optimal moments for goalkeepers to close out on offensive players, research on shot angles provided insight for when goalkeepers can move forward to lower the probability of a shot scoring (Numazu et al., 2024). We are curious about the likelihood of a goal based on the location of a shot (distance and angle) and how much the relationship varies across professional goalkeepers.

Any measurement on the same subject is not useful if it varies substantially between times. In professional soccer, it would be very challenging for scouters if a performance metric varies from season to season. In this regard, we would like to explore whether the save percentage of a goalkeeper is a reliable statistic between two seasons. In other words, we want to determine the strength of the correlation between the save percentages of two consecutive seasons within goalkeepers.

Previous research found that a players' play style has variability through a match. A

player's passing effectiveness, shots, and possession time can change depending on the team's performance, such as close matches versus blowouts. It has been found that players' shots have a higher probability of making the goal when the team is winning (Lago-Peñas, Gómez-Ruano, Megías-Navarro, & Pollard, 2019). In particular, attempting shots inside the box is more likely to score than attempting outside the box. Therefore, we hypothesize that players' decide to attempt shots inside the box more frequently when their team is leading for longer periods.

In addition, age has been observed as a key factor in goalkeeper development. Among junior players, it was found that the oldest goalkeeper in the group is more effective than the youngest by 19%. This finding was not statistically significant, but researchers still believe that goalkeeper skills develop as they age (Obetko, Peráček, Šagát, & Mikulič, 2019). Though it is natural that adult's reaction time becomes slower as they age, we hypothesize that skills and experience outweigh the disadvantage of aging even at the professional level.

We address the four research questions in this project. The first objective is to test and estimate how distance and angle affect the probability of a goal among K League players. The second objective is to assess the reliability of goalkeepers' performance by comparing the save percentages between two seasons. The third objective is to compare the proportion of shots in the box by the proportion of time when a team is leading or trailing. The fourth objective is to evaluate if a goalkeepers' age affects their performance.

2 Methods

2.1 Data Collection

All of the data, through exploring different trends in K1 league, was collected using Fotmob.com. Specifically, we collected the goalkeepers' saved and let through goals in the 2024 and 2025 seasons. Fotmob.com creates shot charts for each goalkeeper, plotting each shot attempted against them throughout the season. The analysis was conducted on nearly every K1 league goalkeeper who faced at least one shot. On the shot charts, circles represent misses and soccer ball icons represent goals, each mapped to its specific location on the field. The objective was to convert these plotted points into XY coordinates, through several extraction methods. The first method was to use generative AI tools such as ChatGPT and Claude by Anthropic, but this approach proved unreliable and inconsistent. For instance, ChatGPT failed to identify partially overlapping circles as separate attempts, detecting only 84 of 120 shots.

The second method, and ultimately the most accurate, was to use a software called "PlotDigitizer." "Plot Digitizer" operates by allowing users to upload an image of a graph and extract data by manually selecting individual points, using reference coordinates to accurately estimate their coordinates. In this case, screenshots of the charts found on Fotmob.com were uploaded, the center of the goal was set at (50,0) and the center line at (50, 100). Then, every individual point was manually clicked using our reference points to specify the XY coordinates.

Subsequently, we merged the datasets and added the goalkeeper, team, and outcome. The outcome is defined as whether the shot was a goal or miss. This process was repeated in creating a dataset for every goalkeeper. The data of 17 goalkeepers in the K1 League from the 2024 season and 29 goalkeepers from the 2025 season were collected, and we could not increase the sample size as there are only 12 teams in the league. With this limited number of goalkeepers, we can generalize

our results to K1 League over these seasons, and if the pattern is similar, it may be generalizable to later seasons. The dataset consists of 5 variables: X-Coordinate, Y-Coordinate, Player, Team, and outcome (shot saved or goal conceded from goalkeeper's perspective). This successfully creates our dataset for objectives one and two.

Continuing with Fotmob.com, additional data were collected to answer objectives three and four. For Objective three, data on FC Seoul's performance throughout the 2025 season was gathered, including the amount of time the team spent leading, trailing, and tied (measured in minutes). Additionally, the number of shots taken from both inside and outside the penalty area were observed. We considered looking at other teams outside FC Seoul but did not have the time to conduct the analysis of those teams as well.

For objective four, data was gathered on the number of shots saved by the goalkeepers, total number of shots attempted on them, their age, and years of experience. Fotmob.com only collects goalkeeper data from 2021 onward, and we gathered data from every goalkeeper observed in the 2024 and 2025 seasons.

2.2 Statistical Methods

2.2.1 First Objective

For objective one, we first considered a logistic regression model which is a generalized linear model for a Bernoulli or binomial outcome. We let Y_i be the binary outcome of each shot on target (make or miss from the goalkeepers perspective). We assumed $Y_i \sim Bern(\theta_i)$ where $\theta_i = \frac{e^{\mu_i}}{1+e^{\mu_i}}$ and $\mu_i = \beta_0 + \beta_1 D_i + \beta_2 A_i$, where D_i and A_i are the distance and angle of the ith shot for $i = 1, 2, ..., n$. Given the (x, y) coordinate of each shot, the distance could be calculated by $\sqrt{(x-50)^2 + y^2}$ and the angle could be calculated by $\arccos\left(\frac{|x_i - 50|}{D_i} \cdot \frac{180}{\pi}\right)$ where (50, 0) represents the center of the goal line.

We found that this model did not describe our observed data well. We found that the angle was not a statistically significant factor as our p-value was greater than our significance level of 0.05. We know that this cannot be the case because angle has a clear affect on the probability of an attempted shot scoring.

We improved the model by using the assumption that the relationship between the probability of a goal and the angle of a shot depends on the shot's distance and we added random-effects to account for the variability between goalkeepers' performance. We let Y_{ij} be the binary outcome of the ith shot on target faced by the jth goalkeeper in the sample. We assumed $Y_{ij} \sim Bern(\theta_{ij})$ where $\theta_{ij} = \frac{e^{\mu}{}_{ij}}{1+e^{\mu}{}_{ij}}$ and $\mu_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})D_{ij} + (\beta_2 + b_{2j})A_{ij} + (\beta_3 + b_{3j})D_{ij}A_{ij}$

In this model, β_0, β_1, β_2, and β_3 are fixed effects, and the parameter of interest was β_3. β_3 is the interaction term of a goal dependent on the angle of the shot and its distance. The player specific random effects b_{0j}, b_{1j}, b_{2j}, b_{3j} are assumed to follow a normal distribution centered at zero. To create this model we will use the Lme4 R package in R as it allows us to create generalized linear mixed-effects models (Bates, Mächler, Bolker, & Walker, 2015). Given the fixed effects, we created a heatmap of the goal at each point on a soccer field across all goalkeepers. Using the random effects, we created player specific heatmaps.

2.2.2 Second Objective

For the second objective, we compared the percentage of shots made on goalkeepers who played in both the 2024 and 2025 K1 League seasons. We expected goalkeepers to remain consistent over the seasons, and created a model to observe whether there was a relationship between each goalkeeper's performance in 2024 and in 2025. A generalized linear model was used to test our hypothesis. We assumed $Y_i \sim Bino(n_i, \theta_i)$ where Y_i is the number of goals saved out of n_i shots faced in 2025. θ_i is the probability of a save and $\theta_i = \frac{e^{\mu_i}}{1+e^{\mu_i}}$. Our model is $\mu_i = \beta_0 + \beta_1 P_{2024,i}$, where $P_{2024,i}$ is the 2024 save percentage for the ith goalkeeper. Given the model, we used ggplot2 (Wickham, 2016), to create a graphic that displayed the proportion of shots allowed in K1 league throughout the 2024 and 2025 seasons.

2.2.3 Third Objective

For objective three, we aimed to understand the difference in shot attempts inside versus outside the penalty box when a team is leading versus trailing. We hypothesize that the findings will support that FC Seoul will shoot inside more when they are leading instead of when they are trailing. To achieve this, we created a generalized linear model where we assumed $Y_i \sim Bino(n_i, \theta_i)$ given Y_i is the number of shots inside the penalty area out of n_i shots attempted. θ_i is the probability of the

shot being inside the penalty area where $\theta_i = \frac{e^{\mu_i}}{1+e^{\mu_i}}$. Our model to predict this is $\mu_i = \beta_0 + \beta_1 T_{Leading}$ with $T_{i,leading}$ indicates whether FC Seoul was leading in the ith game during the 2025 season. After we created a scatterplot using ggplot2 where the y-axis represents the proportion of shots inside while the x-axis represents the time spent leading in minutes.

2.2.4 Fourth Objective

Our fourth objective is to explore how goalkeepers' age affects their performance over time. We hypothesize that performance improves with age as goalkeepers become more skilled. To test this hypothesis, we created a generalized linear model where we assumed $Y_i \sim Bino(n_i, \theta_i)$ given Y_i is the number of goals saved and n_i shots attempted. θ_i is the probability of the goalkeeper saving the shot based on their age where $\theta_i = \frac{e^{\mu_i}}{1+e^{\mu_i}}$. Our model is $\mu_i = \beta_0 + \beta_1 \text{Age}_i$ with Age_i is the age of the ith goalkeeper in the sample. We used ggplot2 to create a graph displaying age versus proportion of goals saved for goalkeepers with at least 3 seasons experience.

0.338999669

3 Results

3.1 First Objective

We considered many different models for assessing how distance and angle affect the probability of a shot scoring on a goalkeeper. We found that the best model was our generalized linear mixed-effects model because it showed that when adding the interaction term between angle and distance, both variables appeared as statistically significant predictors at a significance level of $\alpha = 0.05$. We rejected both $\beta_{distance} = 0$ and $\beta_{angle} = 0$ because distance has a slope of -0.0679 with a p-value = 0.00461 and angle has a slope of 0.0176 with a p-value of 0.01697. Given the two main effects of distance and angle, their interaction was not statistically significant (p = 0.3390). Subsequently, we found the probability of the goalkeeper conceding a goal for every possible point on the soccer field which is mapped from 0-100 from corner to corner for the x-axis and 0-100 from baseline to midfield for the y-axis. This resulted in a heatmap representing the probability of a goalkeeper conceding a goal at every point on the pitch.

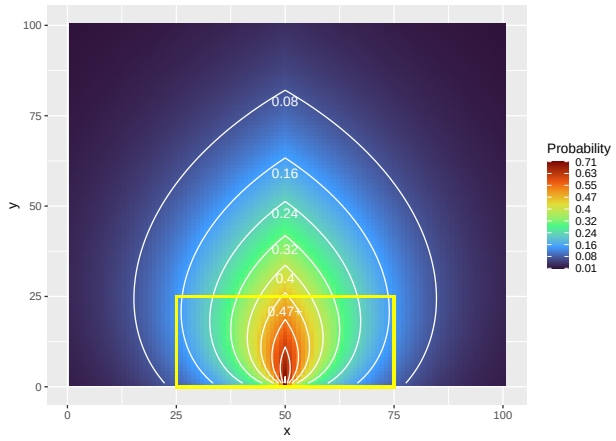


Figure 3.1: Heatmap of Probability of shot going in for average trend across all goalkeepers (fixed-effects only)

Figure 3.1 demonstrates the dramatic increase in probability when the shot attempt is

closer to the goal and non-linear differences between the rings. From the heatmap, we can see a relationship between the distance/angle of the shots and the probability the goalkeeper concedes a goal, as each ring or layer shows a gradual difference in our predicted probabilities. The model estimated that the probability of conceding a goal at an angle of 50 degrees and distance of 0 is 0.5564 while the probability of conceding a goal at an angle of 0 degrees and distance of 50 is 0.0171.

To help teams better understand the importance of a goalkeeper, we can compare goalkeepers to league average and identify areas where their goalkeeper is strong and needs improvement. We wanted to look if we can improve a players performance by understanding their weaknesses. Looking ata particular goalkeeper chosen for demonstration, we identified his most and least proficient areas. First, we created a heatmap using him as our desired player.

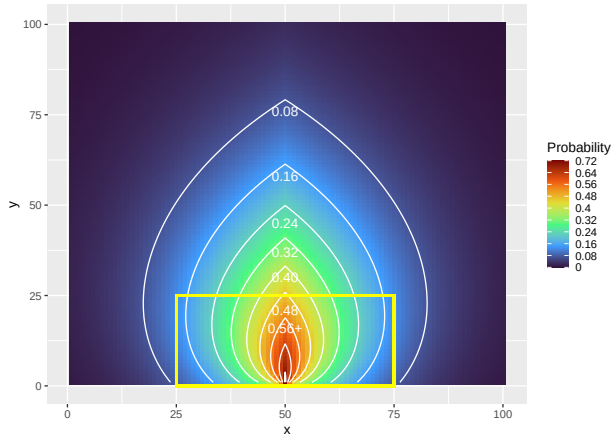


Figure 3.2: Heatmap of Probability of shot going in given a particular goalkeeper chosen for demonstration is the goalkeeper

Since we wanted to see if a particular goalkeeper chosen for demonstration is a better goalkeeper than average, we can use Figure 3.1 and Figure 3.2 to observe the differences between the probability a goal is scored on him versus league average.

In Figure 3.3, we observed that a particular goalkeeper chosen for demonstration is a weaker goalkeeper close to the net than league average because the estimated probabilities of conceding a goal in those regions are higher than the league average. Furthermore, he tends to be stronger than average at medium to far away distances. This allows us to deduce that, at further distances, a particular goalkeeper chosen for demonstration is proficient but he needs improvement at shots within close range. This can be instrumental in the development and training of future goalkeepers.

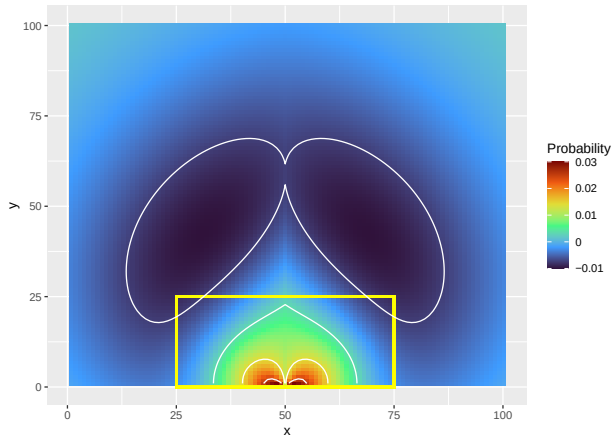


Figure 3.3: Heatmap of Difference in Probability of shot going in given a particular goalkeeper chosen for demonstration is the goalkeeper versus the league average

In general, we can deduce that the angle and distance of the shot matters. The lower the angle and the further the shot, the easier it becomes for the goalkeeper to save a goal.

3.2 Second Objective

To examine differences in player performance between the 2024 and 2025 seasons, we created a model to investigate the relationship between save proportions for goalkeepers who played in K1 League in both seasons. Our model did not find any evidence that goalkeepers' save proportions in 2024 affected their save proportions in 2025 with a z-statistic of 1.706 and a p-value of 0.044. There is an unweighted Pearson's Correlation Coefficient of 0.5788, which tells us that there is a moderately strong linear relationship between the proportions in 2024 and 2025. Figure 3.4 supports this correlation coefficient showing a positive relationship between the proportions in 2024 and 2025.

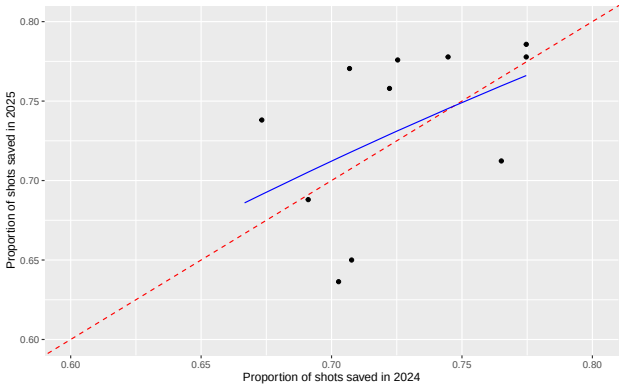


Figure 3.4: Goalies save proportion in the 2024 versus 2025 season

The red line represents a 1 to 1 line where the goalkeepers preformed the same in the 2024 and 2025 seasons. The blue line is using the model to predict the values of y within the range of values that the model was predicting. In general, goalkeepers with high save proportions are more likely to do worse in their next season and vice versa.

3.3 Third Objective

The third objective of this study examines whether a team's game state (leading or trailing) influences shot selection. The difference in time a team spends leading versus trailing is -11.65789 minutes. Using the collected data, we developed a model to assess the relationship between game state and shot attempts. We hypothesized that teams trailing in a match would be more likely to take lower-quality shots, where teams in the lead would take shots from more favorable positions (closer to the goal). Initial analysis of the relationship between these variables did not reveal a statistically significant correlation with a Pearson's Correlation Coefficient of 0.216939 and an observed p-value = 0.158735 which exceeds the significance threshold of $\alpha = 0.05$.

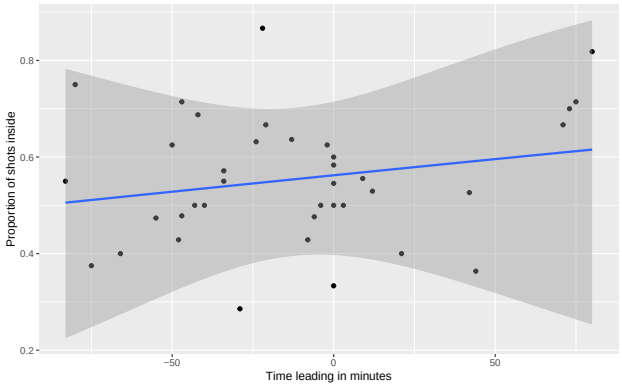


Figure 3.5: Time Leading versus proportion saved

Figure 3.5 displays a clear slight positive relationship between leading time and shot quality. This supports the hypothesis that shot quality is related to whether a team is leading or trailing, but does not provide enough evidence to be conclusive.

3.4 Fourth Objective

Our fourth objective is to analyze whether increasing age affects a players ability to save goals. We first created a generalized linear model which did not find that age has an affect on proportion of saved shots from their years played between 2021 to 2025 (P-value = 0.78948). Since each player has a different trajectory on their careers, it was important to include each individual player as a factor for the model. We expected to find a relationship between a player's age and performance with them improving as they grow older. In our model, this was not observed because the p-value did not support this claim. We used ggplot2 to graph the points of the observed ages against each player's save proportions throughout the years.

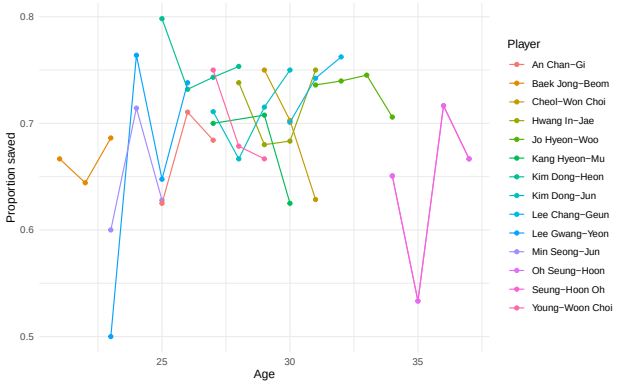


Figure 3.6: Age and Proportion saved over years

Figure 3.6 displays inconsistencies regardless of the player, their age, and their experience. There is no evidence supporting why players improve or worsen throughout the years. With only have three to four year stretches in the data from each player, it is difficult to establish long term trends for each player.

4 Discussion

4.1 Interpreting Results

The results of this study are difficult to interpret due to the limited number of statistically significant findings. Nevertheless, the analysis indicates that shot angle and distance have meaningful effects on the proportion of saves. Within the K1 League dataset, the limited sample size restricted our ability to identify long-term trends. With a larger number of observations, it is likely that stronger evidence would emerge in support of the hypothesized relationships, particularly within objectives three and four. Year-to-year comparisons may provide useful insights into player development by identifying areas on the field where goalkeeping performance can be improved. However, due to the lack of consistent statistical evidence, no definitive conclusions can be drawn regarding systematic changes in goalkeeper performance across seasons.

4.2 Limitations

In our study, each objective has its own limitations. For the first and second objectives, a major limitation is the validity of our points and scale. Even though we scaled the specific images of each of the shot charts, there was no guaranteed way to ensure the shot maps from Fotmob.com was an exact scale from screenshot to screenshot. This could have potentially put points closer or farther away from the goal than the actual distance. These discrepancies may lead to the angle and distance to the goal being incorrect. When generalizing for the whole league, we lose the cases where better players are more likely to make the goal at more difficult ranges and a better goalkeeper can also lower the probability of a goal scoring. As such, it is difficult to standardize how the distance and angle affect the odds of a shot going in.

For objective 3, we only analyzed data from FC Seoul which could make our results

unreliable for other teams in K1 League. Further analysis could be done to better understand how leading and trailing may affect a team's proportion of shots inside versus outside the penalty box. Our Pearson's Correlation Coefficient is also not representative of our model entirely as we could not calculate the exact correlation coefficient for a glm model. For objective 4, if we had access to the players' shot positions and save rates prior to 2021, it could provide more in-depth results on how a player's age affects their performance. Another factor to be considered is the limited playing time a player receives at the start and end of their career. When they age, they receive less playing time and their role is significantly reduced in the off-season. This can misrepresent how a goalkeeper is perceived towards the end of their career. It can also explain why younger goalkeepers start their careers with a much higher save proportion than expected since goalkeepers are either given time to develop in a backup role or in K2 league. Adding data from K2 league can potentially fix the issue by showing goalkeepers' development prior to joining K1 League.

4.3 Challenges and Future Work

A foreseeable challenge is finding a model that correctly fits our data as it may incorrectly predict the odds of a goal going in at different positions. To control all of these limitations and confounding variables, it is best to collect and cross reference our data to better account for goalkeeper skill level and understanding their efficiencies.

This study is effective at putting a small sample into perspective and showing potential trends. As we know from previous studies (Park et al., 2016), variables such as age, distance, and angle affect the results of scoring. This study expands on other works by being able to investigate whether shot charts of different positions on the field can improve training for goalkeepers. This builds on previous studies that have explored goalkeepers' effectiveness in protecting high or low angles of the goal (Numazu et al., 2024), and expands it to a third dimension which includes defending different regions of the pitch. Our study also shows how volatile goalkeepers' save percentages are based on team success and that their save percentages are valuable statistics.

In the future, we can study additional ways to determine how valuable a teams' goalkeeper is to their own team compared to other goalkeepers in the league. Expanding the study to other leagues, such as Premier league and LaLiga, can help us further understand whether these trends

exist in other leagues. We could potentially bootstrap the data across goalkeepers to explore greater differences among objectives three and four. We could also seek more precise and accurate ways to collect the data.

5 Conclusion

We found that angle and distance are significant predictors of the probability of saving an on-goal shot, and that data visualizations, such as heatmaps, can represent these results more effectively. By comparing an individual goalkeeper to league average, we can understand each player's individual strengths and weaknesses. Furthermore, the heatmaps, can help offensive players effectively find better placements of shots on goal and increasing their probability of scoring. Although we did not observe strong correlation coefficients or p-values for our models, we still observed trends in time spent leading versus proportion of shots inside. We also found a weak correlation between age and shot proportion. A combination of these studies suggest it can be unreliable to analyze goalkeepers or teams performance consistently.

References

Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, *67*(1), 1–48. doi: 10.18637/jss.v067.i01

Christiansen, A. T., & Houser, B. T. (2024). Analyzing goalkeeper impact in premier league football. *Proceedings of the Western Users of SAS Software (WUSS)*, *2024*, Paper 182. https://www.wuss.org/proceedings/2024/182/182_FINAL_paper_pdf.pdf

Lago-Peñas, C., Gómez-Ruano, M., Megías-Navarro, D., & Pollard, R. (2019). Exploring elite soccer teams' performances during different match-status periods. *Information Sciences*, *499*, 137–147. https://www.sciencedirect.com/science/article/pii/S0960077919305235 doi: 10.1016/j.ins.2019.05.023

Numazu, N., et al. (2024). Analysis of soccer goalkeeper performance and shot scenarios in the 2022 world cup. *Journal of Physical Education and Sport*. https://www.researchgate.net/publication/387386037

Obetko, M., Peráček, P., Šagát, P., & Mikulič, M. (2019). Impact of age and agility performance level on the disjunctive reaction time of soccer goalkeepers. *Acta Facultatis Educationis Physicae Universitatis Comenianae*, *59*(2), 224–238. https://www.researchgate.net/publication/337451995_Impact_of_Age_and_Agility_Performance_Level_on_the_Disjunctive_Reaction_Time_of_Soccer_Goalkeepers doi: 10.2478/afepuc-2019-0020

Park, Y.-S., Choi, M.-S., Bang, S.-Y., & Park, J.-K. (2016). Analysis of shots on target and goals scored in soccer matches: Implications for coaching and training goalkeepers. *South African Journal for Research in Sport, Physical Education and Recreation*, *38*(1), 123-137.

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York. doi: 10.1007/978-3-319-24277-4